Wind and Solar Radiation Complementarity Analysis

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Abstract

This study examines the temporal and geographical patterns of complementarity between wind speed and solar radiation in Switzerland. Using hourly observations from 129 MeteoSwiss stations between 2020 and 2023, I calculate Pearson and Spearman correlations at hourly, daily, and monthly levels. Correlations in the full sample are generally weak, while restricting the analysis to observations above one and/or both potential-production thresholds yields negative correlations at each time scale. Subgroup results also suggest variation by season, location, and elevation. These findings indicate that observed complementarity depends on the time scale and sample definition: they describe meteorological associations rather than electricity-production profiles.

Keywords: wind; solar radiation; complementarity; Switzerland.
JEL classification: to be completed

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Introduction

I explore the geographical and temporal conditions under which wind speed and solar radiation are complementary in Switzerland.

I collect wind and solar radiation data from MeteoSwiss at hourly, daily and monthly levels, including information on time perdiod, location and elevation. These variables are of particular interest because they form the basis of wind and solar power generation. This focus on the meteorological determinants of renewable electricity generation provides a direct link to the analysis conducted by Cauz et al. [2025], who explored the complementarity between electricity production from wind turbines and photovoltaic (PV) panels and found a negative correlation between the two, thereby mitigating the risks to which these renewable energy sources are exposed.

Overall, I find evidence of wind–solar complementarity in Switzerland, depending on the time horizon, time period, and location. The negative correlation is stronger in subsamples corresponding to areas with high potential for electricity production, characterized by strong winds and/or high solar radiation.

Moreover, wind–solar complementarity is observed across the country, but it is stronger during winter, in the northern part of Switzerland, and at elevations above 800 meters, with particularly strong complementarity above 1,200 meters. Negative correlations are also observed at lower elevations, below 400 meters.

Across hourly, daily, and monthly observations, Spearman correlations are 0.1333, 0.0772 and 0.0938 in the full sample, -0.0965, -0.1578 and -0.1477 in the subsample where measures are both above the thresholds and -0.4161, -0.2822, -0.2062 in the second subsample where at least one out of two measures is above its threshold (Table 6, Table 5, Table 7). These results suggest a potential complementarity between solar panels and wind tubines in high potential areas. The thresholds-restricted estimates are more negative, but they describe associations between meteorological observations, not electricity production or guaranteed electricity supply complementarity.

The paper first describes the data and measures of complementarity (section 1 and section 2), then presents the results (section 3), discusses sensitivity and limitations (subsection 3.1 and section 4), and concludes (section 5).

1 Data and sample construction

I use data from MeteoSwiss from 2020 to 2023[2], relying on measurements of mean wind speed (in m/s) and solar radiation (in W/m^2) at hourly, daily, and monthly levels across Switzerland. The choice of these variables constitutes one of the hypotheses underlying this work, which I discuss and justify in another document. In short, electricity generation from wind turbines depends on wind speed, while solar radiation drives electricity generation from photovoltaic (PV) systems.

The unit of observation is a station-hour, station-day or station-month, with wind speed and solar radiation observed jointly. At the hourly, daily and monthly levels,

[2]I selected this period to align with the work of Cauz et al. [2025], which provides a benchmark.

respectively, the full sample (*FULL*) contains 4,453,593, 185,736 and 6,114 paired observations. The *AND* subsample, which retains observations with wind speed ≥ 3 m/s **and** solar radiation ≥ 100 W/m^2, contains 604,689, 32,108 and 1,222 observations. The *OR* subsample, which retains observations with wind speed > 3 m/s or solar radiation > 100 W/m^2 (or both), contains 2,331,485, 137,539 and 4,686 observations respectively. Unless otherwise indicated, the tables presenting my results below report my own calculations using these MeteoSwiss data for 2020–2023. Comparison estimates attributed to Cauz et al. [2025] are taken from their study.

The (*FULL*) sample covers 129 stations across Switzerland. The *AND* subsample retains 126 stations at the daily level and 71 at the monthly level while the *OR* subsample retains all 129 stations at both the daily and monthly levels too.

Stations are grouped into four elevation classes: <400 m, 400–<800 m, 800–<1,200 m, and ≥1,200 m and are defined in four geographical zones with no administrative meaning between North, South, East and West.

Below, I display a map showing station locations across Switzerland, divided into four areas.



Figure 1: Geographical distribution of Swiss Weather Stations

An important feature of wind and solar energy is the existence of minimum production thresholds. When wind speed or solar radiation is too low, the corresponding technology cannot generate electricity. I use thresholds of 3 m/s for wind speed and 100 W/m^2 for solar radiation to identify conditions under which electricity generation is potentially feasible. Actual generation depends on the specific wind turbine or PV system considered, so results based on these thresholds should be interpreted as conditional on these assumptions.

Accordingly, throughout the rest of the analysis, I compare, whenever relevant, the

full sample (*FULL*), which imposes no threshold, with two subsamples. The first (*AND*) includes only observations for which both thresholds are exceeded (wind speed above 3 m/s ****and**** solar radiation above 100 W/m^2). The second (*OR*) includes observations for which at least one of the two variables exceeds its respective threshold (wind speed above 3 m/s ****or**** solar radiation above 100 W/m^2).

Across paired station-days from 2020 to 2023, 29.4% of mean wind speed observations in the *FULL* sample exceed 3 m/s, while 61.9% of mean solar radiation observations exceed 100 W/m^2. The corresponding figures for the *OR* sample are 39.7% and 83.6%. Appendix C presents descriptive statistics for both measures across time, geographical zones, and elevation.

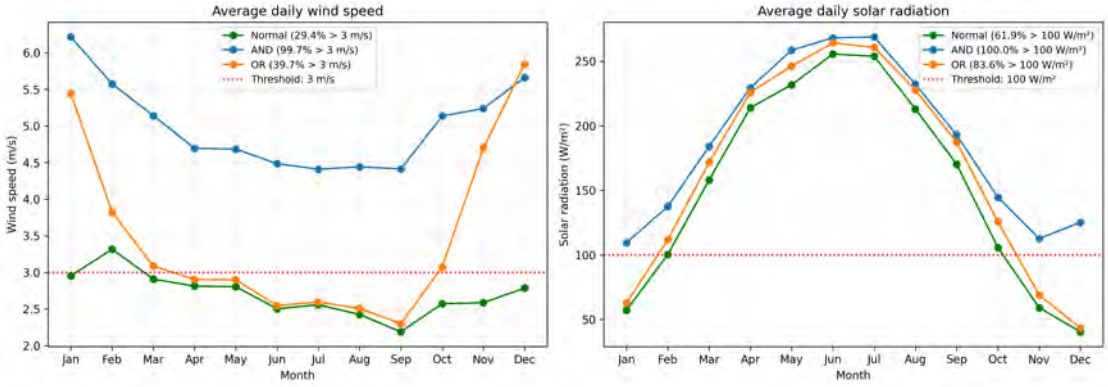


Figure 2: Monthly profiles of mean daily wind speed and solar radiation, 2020–2023.
Notes: Legend percentages give the share of station-days in each sample strictly above the relevant threshold over the full period.
Source: Own calculations using MeteoSwiss data.

2 Measuring resource complementarity

Using hourly, daily and monthly data, I compute Spearman and Pearson correlations to explore potential complementarity between the two resources. Pearson correlation measures linear association, whereas Spearman correlation measures monotonic association based on ranks and is a preferred method in this type of analysis, as noted by Cauz et al. [2025].

We distinguish here between data at hourly, daily and monthly levels. Hourly measures identify very short-term variations, making it possible to study complementarity within a day. Daily data identify variations between days (one day may be cloudy and windy, while another is clear with low wind). Whereas, monthly data attenuate short-term variation and identify seasonal profiles.

I distinguish between seasonal profiles, since solar radiation tends to be lower during winter due to fewer daylight hours and greater cloud cover. I also examine simultaneous observations of wind speed and solar radiation at the same station and hour using

Spearman and Pearson correlations. Finally, I explore how frequently either resource, or both resources simultaneously, exceed their respective potential production thresholds.

The *AND* subsample is used to explore complementarity when both wind speed and solar radiation are above their respective thresholds, allowing us to examine how the two variables vary within a subsample characterized by conditions potentially suitable for electricity generation. By contrast, the *OR* subsample captures cases in which at least one of the two variables is above its respective threshold. It therefore allows us, for instance, to study complementarity at night, or when wind speed is below its threshold but solar radiation remains sufficiently high, without excluding periods during which both variables are above their respective thresholds.

Thresholds are applied separately at each temporal resolution. For daily and monthly analyses, I first calculate station-level averages using all valid hourly wind–radiation pairs, then apply the *AND* or *OR* filter to these averages. Consequently, low solar radiation at night does not automatically exclude a day: inclusion depends on the daily averages and the chosen filter. For example, some hours may satisfy both thresholds even if the daily averages do not. Conversely, a day may meet both thresholds on average despite containing hours below them. The same principle applies to monthly observations.

3 Results: overall complementarity

My findings are summarised in Table 1. Additional correlations stratified by month, geography and elevation are reported in Appendix A. Within-season daily correlations are presented in Table 3.

Table 1: Overall wind speed–solar radiation correlations by sample and temporal resolution, 2020–2023.

Sample	Level	N	Pearson	Spearman
FULL	Hourly	4,453,593	0.0925	0.1333
	Daily	185,736	-0.0459	0.0772
	Monthly	6,114	-0.0180	0.0938
AND	Hourly	604,689	-0.1258	-0.0965
	Daily	32,108	-0.1713	-0.1578
	Monthly	1,222	-0.2392	-0.1477
OR	Hourly	2,331,485	-0.3119	-0.4161
	Daily	137,539	-0.3405	-0.2822
	Monthly	4,686	-0.3334	-0.2062

Notes: N counts paired wind–solar radiation observations, measured in station-hours, station-days or station-months. FULL includes all valid pairs; AND requires wind speed $\geq$ 3 m/s and solar radiation $\geq$ 100 W/m^2; OR requires wind speed $>$ 3 m/s or solar radiation $>$ 100 W/m^2 (or both).
Source: Own calculations using MeteoSwiss data, 2020–2023.

The comparison with Cauz et al. [2025], whose estimates are reported in Table 4 in the appendix, is indicative rather than directly like-for-like: their estimates use national

electricity-generation profiles, whereas mine use pooled station-level meteorological observations, which differ in unit of observation and normalisation.

Without thresholds, the Pearson correlation is positive at the hourly level and slightly negative at the daily and monthly levels, while the Spearman correlation is always positive but weak. In both filtered subsamples, the Pearson and Spearman correlations are negative. For the whole dataset (*FULL*), we remain cautious when it comes to complementarity, since Spearman correlations are not negative at any of these levels.

As in Cauz et al. [2025, Figure 1], I divide hourly wind speed and solar radiation observations into four zones using the thresholds of 3 m/s and 100 W/m^2. For this classification, a resource is considered high when its measure is at or above its threshold. Figure 3 and Figure 4 compare the *FULL* and *OR* samples on the same axes. In the *FULL* sample, 18.8% of paired observations meet only the wind threshold, 20.7% meet only the solar radiation threshold, 13.6% meet both, and 46.9% meet neither.

Within the *OR* sample (N = 2,331,485 station-hours), the corresponding shares are 34.6% for wind alone, 39.5% for solar radiation alone, 25.9% for both, and 0% for neither. These percentages are conditional on *OR* selection, which requires wind speed $>$ 3 m/s or solar radiation $>$ 100 W/m^2. The absence of observations below both thresholds follows directly from this selection rule.

3.1 Results: seasonal profiles and within-season correlations

The monthly profiles in Figure 2 show opposing seasonal patterns in mean wind speed and solar radiation. Table 2 summarises daily observations for *FULL*, *AND* and *OR*, defining winter as December–February, spring as March–May, summer as June–August, and autumn as September–November. Each season pools observations from 2020 to 2023.

In the *FULL* sample, daily mean wind speed averages 3.011 m/s in winter and 2.497 m/s in summer. Mean daily solar radiation averages 65.01 W/m^2 in winter and 240.65 W/m^2 in summer. In all samples, winter has higher mean wind speed and lower mean solar radiation than summer. These comparisons suggest complementarity between seasonal resource profiles.

Opposing seasonal averages do not establish a negative association within each season. I therefore also calculate Pearson and Spearman correlations using paired station-days separately for each season and sample (Table 3). In the *AND* sample, both correlations are negative in all four seasons, but they are least negative in winter (Pearson: -0.0431; Spearman: -0.0372) and most negative in autumn (-0.1756 and -0.1815, respectively). In *FULL*, within-season correlations remain weak. In *OR*, the Spearman correlation ranges from -0.4539 in winter to 0.0156 in summer.

The sign also depends on temporal resolution. For monthly *AND* observations in winter, Pearson is 0.0588 and Spearman is 0.1435 (N = 108 station-months). Thus, the daily results do not establish negative correlations at every time scale, and seasonal mean differences should be distinguished from associations within seasons.

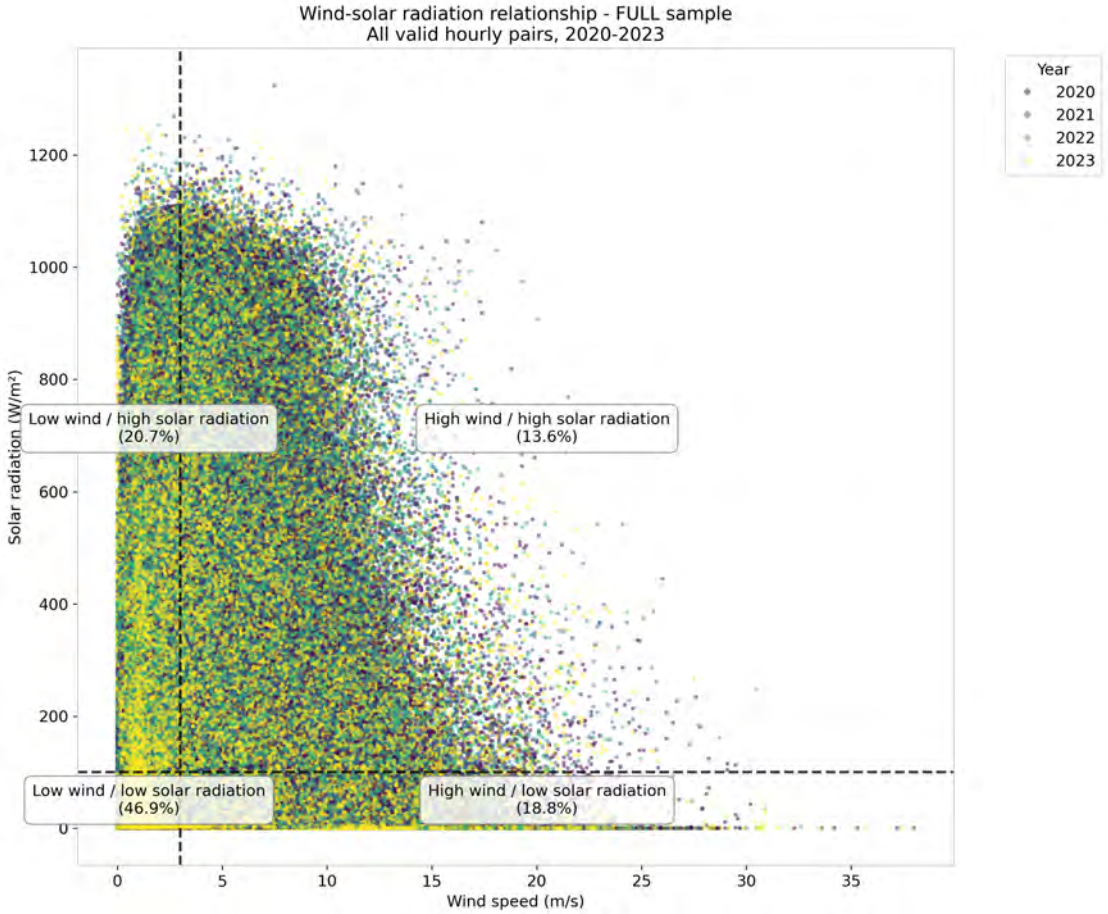


Figure 3: Hourly wind speed and solar radiation in the *FULL* sample (N = 4,453,593 station-hours), 2020–2023. Dashed lines indicate the thresholds of 3 m/s and 100 W/m^2.
Source: Own calculations using MeteoSwiss data.

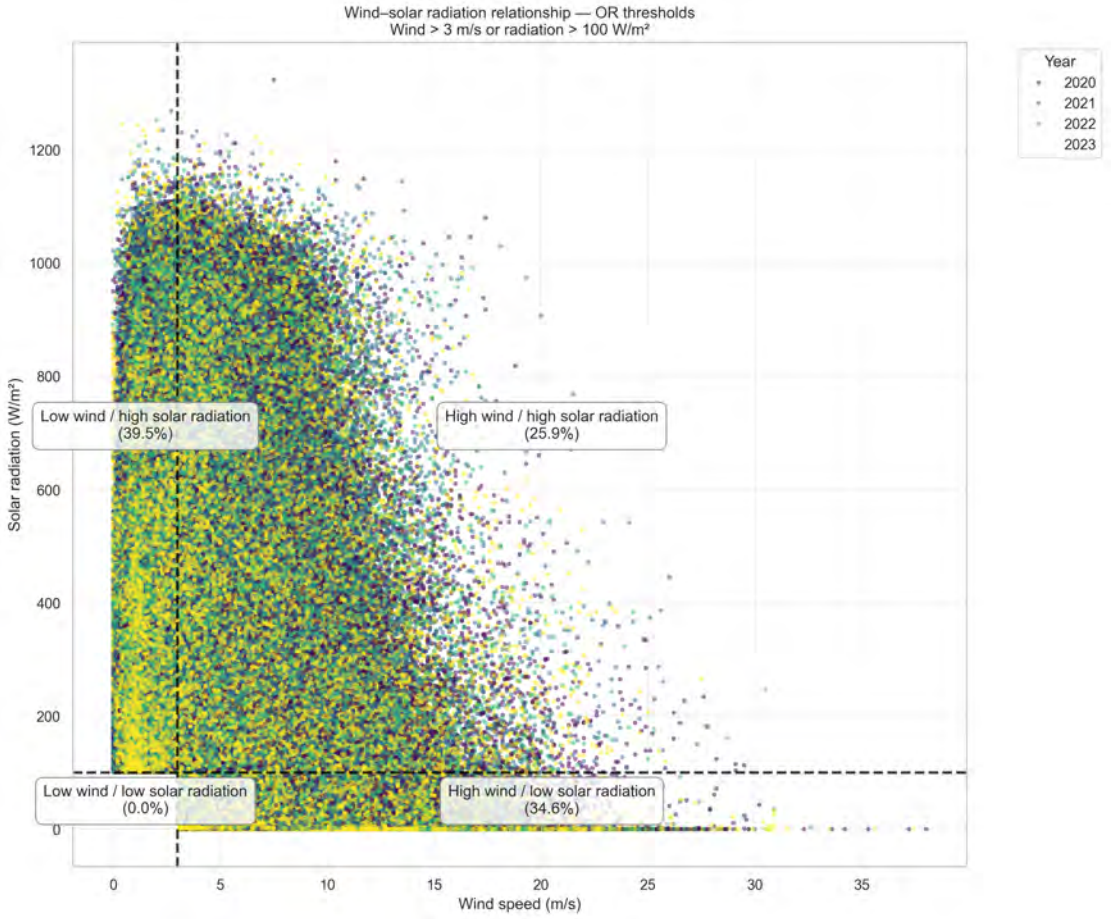


Figure 4: Hourly wind speed and solar radiation in the *OR* sample (N = 2,331,485 station-hours), 2020–2023. Percentages are conditional on OR selection; the empty low-wind, low-radiation zone follows from the filter.

Source: Own calculations using MeteoSwiss data.

Table 2: Seasonal descriptive statistics from daily observations, 2020–2023.

			Wind speed (m/s)		Solar radiation (W/m^2)	
Sample	**Season**	**N**	**Mean**	**Median**	**Mean**	**Median**
FULL	Winter	45,923	3.011	2.050	65.01	57.83
	Spring	46,792	2.844	2.275	201.12	204.29
	Summer	46,753	2.497	2.079	240.65	255.00
	Autumn	46,268	2.453	1.742	111.71	98.75
AND	Winter	3,278	5.686	4.954	132.63	127.83
	Spring	12,929	4.826	4.217	226.14	222.17
	Summer	10,821	4.444	3.954	257.36	265.08
	Autumn	5,080	4.826	4.221	162.51	153.29
OR	Winter	21,768	4.714	4.117	83.30	82.12
	Spring	42,358	2.963	2.375	215.23	215.75
	Summer	44,064	2.552	2.121	251.12	261.75
	Autumn	29,349	3.024	2.225	142.95	141.46

Notes: N counts paired station-days. Winter: December–February; spring: March–May; summer: June–August; autumn: September–November. Seasons pool all years from 2020 to 2023. Daily means are calculated from valid hourly pairs before filtering.
Source: Own calculations using MeteoSwiss data.

Table 3: Within-season wind speed–solar radiation correlations from daily observations, 2020–2023.

Sample	**Season**	**N**	**Pearson**	**Spearman**
FULL	Winter	45,923	0.0296	0.0539
	Spring	46,792	-0.0160	0.0578
	Summer	46,753	0.0152	0.0854
	Autumn	46,268	-0.0936	-0.0136
AND	Winter	3,278	-0.0431	-0.0372
	Spring	12,929	-0.1086	-0.0879
	Summer	10,821	-0.1029	-0.0896
	Autumn	5,080	-0.1756	-0.1815
OR	Winter	21,768	-0.3805	-0.4539
	Spring	42,358	-0.1402	-0.0581
	Summer	44,064	-0.0704	0.0156
	Autumn	29,349	-0.4573	-0.4405

Notes: Correlations pool station-days within each season and sample. N counts complete wind–radiation pairs; filters are applied to daily means. Season definitions are given in Table 2.
Source: Own calculations using MeteoSwiss data.

3.2 Results: altitude and geographical heterogeneity

Descriptive statistics show that hourly mean wind speed generally increases across elevation classes in the *FULL* and *OR* samples, whereas the pattern is not strictly monotonic in *AND*. Mean solar radiation also varies across elevation classes and sample. These are differences in averages, not isolated effects of elevation or cloud cover. In particular, a higher mean wind speed does not by itself imply a more negative wind–solar correlation (Table 12).

Figure 5 shows the monthly profiles of mean daily wind speed by elevation class for the *FULL*, *AND* and *OR* samples. The profiles illustrate both the higher average wind speeds at many higher-elevation stations and the variation across months and sample definitions.

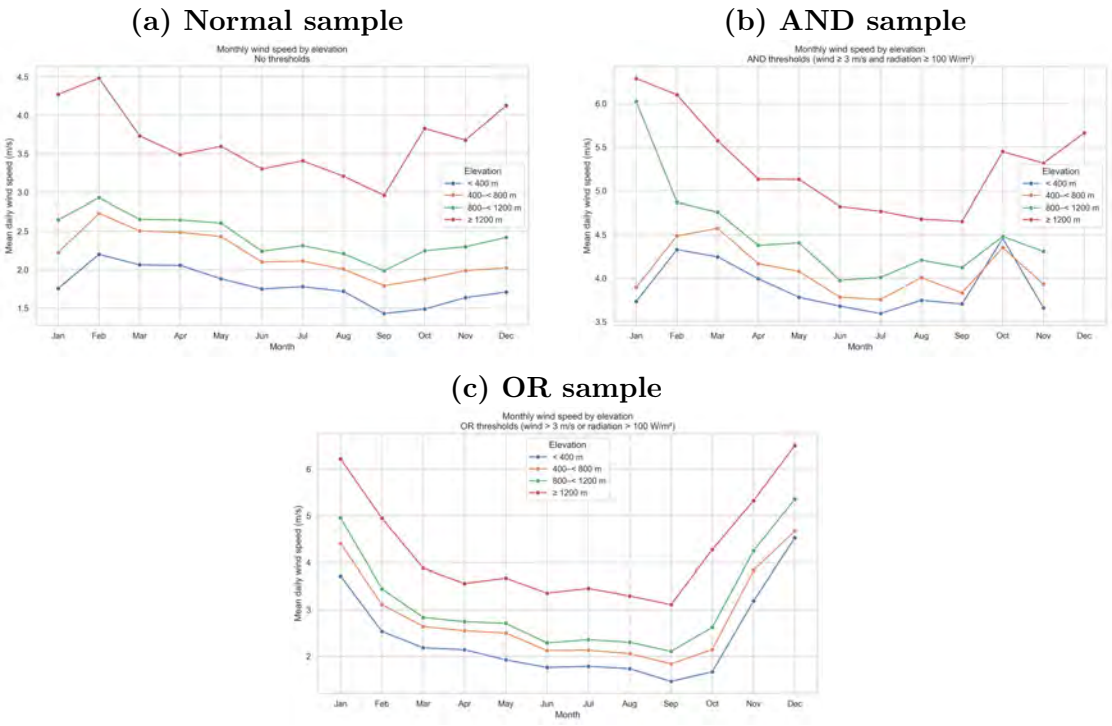


Figure 5: Monthly mean daily wind speed by elevation class and sample definition.

The relationship between wind speed and solar radiation also varies by geographical zone and elevation. The hourly correlation heatmaps in Figure 6 report Spearman correlations in colour and as the top number in each cell, while Pearson correlations appear between brackets. Their common colour scale facilitates comparisons across the *FULL*, *AND* and *OR* samples. Blank cells indicate that a correlation is unavailable.

In the *FULL* sample, the correlations differ in sign across locations: for example, the Spearman correlation is negative in the North at $\geq$ 1,200 m (-0.280), but positive in the East at the same elevation (0.055). Under the *AND* sample, the most negative Spearman correlation is in the North at 800–$<$ 1,200 m (-0.284). Under *OR*, the strongest negative

value is likewise in the North at 800–< 1,200 m (−0.562), compared with −0.463 in the North at ≥ 1,200 m. The North and West also show negative correlations in several other elevation classes. These subgroup correlations are descriptive and do not identify causal effects or rank potential production sites.

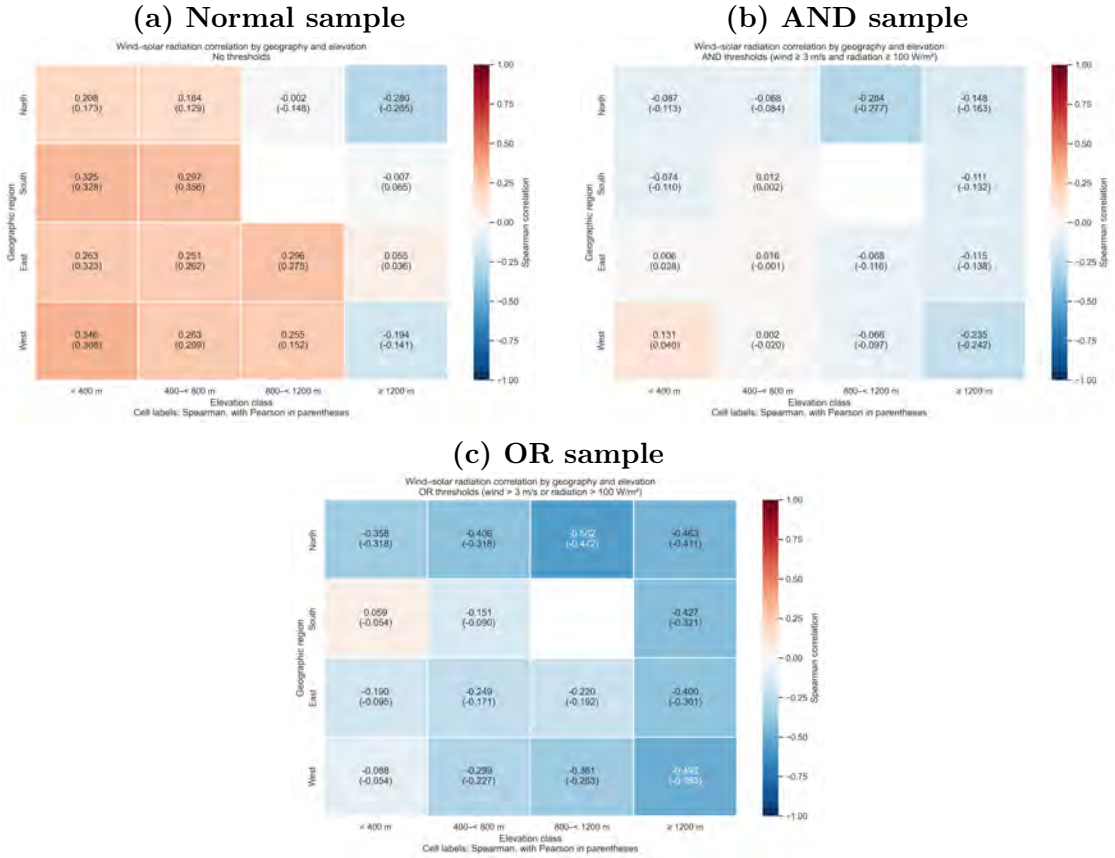


Figure 6: Hourly wind–solar correlations by geographical zone, elevation class and sample definition. Cell colours and the first number show Spearman correlations; Pearson correlations are in parentheses. All panels use a common colour scale. Empty cells indicate unavailable correlations.

4 Sensitivity analysis and limitations

Overall, results are sensitive to thresholds and the method used to compute correlations. Analyses with thresholds yield stronger negative correlations but examine meteorological observations with the potential to support electricity production. Pearson and Spearman correlations distinguish between linear and monotonic associations without establishing complementarity.

Fixed effects regressions distinguish between sources of variation. Station fixed effects (FEs) control for time-invariant mean differences across station locations, while time FEs

control for variation common to all stations in each observed period. Thus, estimated coefficients describe a conditional relation, not a causal one. A change in sign after the introduction of FE suggests common variation across stations.

For each temporal resolution, I regress solar radiation on wind speed using a simple specification, a station fixed-effects specification, a time fixed-effects specification, and a specification including both sets of fixed effects. The wind-speed coefficient measures the change in solar radiation associated with a one-m/s increase in wind speed; standard errors are two-way clustered by station and time. Regression results are reported in Table 8 and Table 9.

In addition, correlations within station groups encompass variation that is not visible at the aggregate level. That is why it is important to distinguish stations by altitude and location. Estimates for small subgroups, particularly at the monthly level, should be interpreted cautiously. For example, the threshold-restricted winter group below 400 m contains only two observations, making its correlation estimate uninformative about a general relationship.

Additional checks remain necessary concerning variation in thresholds, the final number of observations, especially for monthly measures, and the years studied, *etc.*

We should bear in mind that we are studying complementarity between meteorological resources, not between electricity production profiles. Thus, we need to remain cautious when drawing conclusions about complementarity between wind turbines and PV installations.

5 Conclusion

This study finds opposing seasonal profiles for the underlying meteorological resources: daily mean wind speed is higher in winter, while mean solar radiation is higher in summer (Table 2). This pattern is consistent across the FULL, AND and OR samples and suggests seasonal resource complementarity, but it does not imply a negative association at every time scale or within every season. In the FULL sample, overall Spearman correlations are weakly positive at hourly, daily and monthly levels (0.1333, 0.0772 and 0.0938). They are negative in AND (-0.0965, -0.1578 and -0.1477) and more negative in OR (-0.4161, -0.2822 and -0.2062), respectively (Table 5, Table 6, Table 7). The strength of the observed association therefore depends on sample selection as well as temporal resolution.

Seasonal correlations further qualify this result: within-season associations in FULL are generally weak, those in AND are negative across the four seasons, and those in OR are most negative in winter and autumn, while its summer Spearman correlation is close to zero and slightly positive (Table 3). Geographic and elevation patterns are likewise not uniform. For example, the strongest negative hourly Spearman correlation shown in the OR heatmap occurs in the North at 800–$<$ 1,200 m (-0.562), rather than in the highest elevation class; the unfiltered heatmap also contains positive correlations in some groups (Figure 6). These subgroup patterns are descriptive and conditional on the selected observations, not evidence that a particular elevation or region guarantees

complementarity.

Finally, this analysis measures associations between meteorological wind speed and solar radiation, not between electricity-generation profiles. The results do not establish the magnitude, reliability or contribution to demand coverage of wind–solar electricity production. Future work should translate these resources into potential generation using appropriate technology and site characteristics, then assess the resulting production profiles against demand.

A Additional correlation results

A.1 Published benchmark

Table 4: Spearman correlations between wind and solar electricity production reported by Cauz et al. (2025).

Level	Spearman
Hourly	-0.176
Daily	-0.368
Monthly	-0.736

Source: Cauz et al. [2025].

A.2 Hourly stratified correlations

Table 5: Hourly wind speed–solar radiation correlations: no thresholds, AND and OR samples. N counts complete wind–radiation pairs. Calendar months pool all years.

	Threshold	No thresholds			AND			OR		
	Statistic	N	Pearson	Spearman	N	Pearson	Spearman	N	Pearson	Spearman
Breakdown	Group									
Overall	All observations	4,453,593	0.0925	0.1333	604,689	-0.1258	-0.0965	2,331,485	-0.3119	-0.4161
Month	01 Jan	377,643	0.0347	-0.0191	28,092	-0.0135	-0.0020	172,554	-0.3827	-0.4875
	02 Feb	344,075	0.0079	0.0073	41,120	-0.0885	-0.0845	188,965	-0.3809	-0.4830
	03 Mar	378,137	0.0812	0.1268	58,626	-0.0992	-0.0836	210,773	-0.3363	-0.4239
	04 Apr	366,138	0.1589	0.2131	72,790	-0.0546	-0.0366	214,590	-0.2522	-0.3101
	05 May	378,099	0.1993	0.2629	81,312	-0.0356	-0.0082	227,956	-0.1772	-0.2289
	06 Jun	366,033	0.2048	0.2726	68,336	-0.0548	-0.0296	217,416	-0.1788	-0.1979
	07 Jul	377,541	0.2046	0.2677	71,992	-0.0503	-0.0259	225,094	-0.1850	-0.1988
	08 Aug	377,650	0.1774	0.2304	59,635	-0.0421	-0.0191	209,713	-0.2342	-0.2668
	09 Sep	365,275	0.1440	0.1885	42,468	-0.0840	-0.0571	182,620	-0.3114	-0.3347
	10 Oct	378,540	0.0528	0.0775	37,039	-0.1185	-0.0998	178,530	-0.4010	-0.5016
	11 Nov	365,649	0.0290	0.0015	24,153	-0.0672	-0.0531	152,065	-0.4310	-0.5446
	12 Dec	378,813	0.0138	-0.0539	19,126	-0.0479	-0.0366	151,209	-0.4125	-0.5065
Altitude	<400 m	490,610	0.2513	0.2548	43,634	-0.0344	-0.0302	202,984	-0.1974	-0.2225
	400–<800 m	1,794,627	0.2016	0.2413	205,930	-0.0124	0.0033	825,872	-0.2340	-0.3155
	800–<1200 m	595,071	0.1238	0.2118	75,620	-0.1497	-0.1193	298,581	-0.3243	-0.4209

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Table 5: Hourly wind speed–solar radiation correlations: no thresholds, AND and OR samples. N counts complete wind–radiation pairs. Calendar months pool all years.

	Threshold	No thresholds			AND			OR		
	Statistic	N	Pearson	Spearman	N	Pearson	Spearman	N	Pearson	Spearman
Breakdown	Group									
	≥1200 m	1,573,285	-0.0111	-0.0304	279,505	-0.1753	-0.1543	1,004,048	-0.3376	-0.4474
Geography	North	919,029	0.0638	0.1222	96,573	-0.1400	-0.1219	461,934	-0.3817	-0.4805
	South	665,049	0.1244	0.1042	83,577	-0.1092	-0.0917	335,988	-0.2924	-0.3786
	East	1,364,806	0.1183	0.1412	210,337	-0.1319	-0.1054	725,717	-0.2835	-0.3832
	West	1,504,709	0.0691	0.1446	214,202	-0.1477	-0.1053	807,846	-0.3234	-0.4283

A.3 Daily stratified correlations

Table 6: Daily wind speed–solar radiation correlations: no thresholds, AND and OR samples. N counts complete wind–radiation pairs. Calendar months pool all years.

	Threshold	No thresholds			AND			OR		
	Statistic	N	Pearson	Spearman	N	Pearson	Spearman	N	Pearson	Spearman
Breakdown	Group									
Overall	All observations	185,736	-0.0459	0.0772	32,108	-0.1713	-0.1578	137,539	-0.3405	-0.2822
Month	01 Jan	15,755	0.0623	0.0433	573	-0.0025	0.0125	5,860	-0.1051	-0.1088
	02 Feb	14,350	-0.1320	-0.1200	2,698	-0.0003	0.0142	11,016	-0.3703	-0.3625
	03 Mar	15,768	-0.0777	-0.0452	3,907	-0.0375	-0.0254	13,751	-0.2525	-0.1918
	04 Apr	15,262	-0.0231	0.0310	4,395	-0.0689	-0.0344	14,124	-0.1516	-0.0750
	05 May	15,762	0.0656	0.1454	4,627	-0.1030	-0.0729	14,483	-0.0380	0.0523
	06 Jun	15,259	0.0289	0.0994	3,485	-0.1155	-0.1006	14,566	-0.0394	0.0425
	07 Jul	15,744	0.0136	0.0863	4,027	-0.1101	-0.0799	15,171	-0.0443	0.0393
	08 Aug	15,750	-0.0200	0.0362	3,309	-0.0868	-0.0837	14,327	-0.1568	-0.0691
	09 Sep	15,238	-0.0849	-0.0065	2,257	-0.0993	-0.0912	13,110	-0.2752	-0.1362
	10 Oct	15,783	-0.0728	-0.0083	2,246	-0.0800	-0.0654	10,917	-0.4096	-0.3094
	11 Nov	15,247	0.0049	0.0255	577	0.0079	0.0025	5,322	-0.3289	-0.3747
	12 Dec	15,818	0.0609	0.0597	7	-0.1218	-0.1071	4,892	-0.0324	0.0195
Altitude	<400 m	20,449	0.0069	0.1187	1,180	-0.1448	-0.1340	13,492	-0.3176	-0.1480
	400–<800 m	74,829	-0.0199	0.0636	7,709	-0.1182	-0.1088	51,467	-0.3415	-0.2414
	800–<1200 m	24,822	-0.0421	0.0901	3,255	-0.0998	-0.0848	17,919	-0.3722	-0.2610
	≥1200 m	65,636	-0.1319	-0.0169	19,964	-0.2007	-0.1938	54,661	-0.3637	-0.3403
Geography	North	38,323	-0.1430	-0.0722	4,271	-0.1622	-0.1517	27,012	-0.4554	-0.3872
	South	27,730	0.0348	0.1943	4,806	-0.1552	-0.1340	20,607	-0.2859	-0.1879
	East	56,931	0.0187	0.1780	12,563	-0.2070	-0.1982	42,513	-0.2901	-0.2228
	West	62,752	-0.0926	0.0195	10,468	-0.1601	-0.1410	47,407	-0.3613	-0.3100

A.4 Monthly stratified correlations

Table 8: Regression results for the wind–solar relationship

Temporal scale	Model	Coefficient	Standard error	p-value	Observations (N)
1. Hourly	Simple	9.0325	2.3380	1.1187e-04	4,453,593
	Station FE	11.4943	3.0639	1.7575e-04	4,453,593
	Time FE	2.5051	0.3366	9.8810e-14	4,453,593
	Station + time FE	1.9891	0.3948	4.7044e-07	4,453,593
2. Daily	Simple	-2.2008	0.7790	4.7261e-03	185,736
	Station FE	-6.8185	1.3843	8.4214e-07	185,736
	Time FE	2.6294	0.4566	8.4636e-09	185,736
	Station + time FE	1.7456	0.4989	4.6727e-04	185,736
3. Monthly	Simple	-0.9140	1.8107	6.1371e-01	6,114
	Station FE	-18.1413	7.2374	1.2217e-02	6,114
	Time FE	3.0738	0.7121	1.6095e-05	6,114
	Station + time FE	1.1248	0.9513	2.3711e-01	6,114

Table 7: Monthly wind speed–solar radiation correlations: no thresholds, AND and OR samples. N counts complete wind–radiation pairs. Calendar months pool all years.

	Threshold	No thresholds			AND			OR		
	Statistic	N	Pearson	Spearman	N	Pearson	Spearman	N	Pearson	Spearman
Breakdown	Group									
Overall	All observations	6,114	-0.0180	0.0938	1,222	-0.2392	-0.1477	4,686	-0.3334	-0.2062
Month	01 Jan	509	0.3233	0.1878	0	–	–	169	0.3263	0.3683
	02 Feb	509	0.2408	0.1721	108	0.0588	0.1435	317	-0.0254	-0.0581
	03 Mar	509	0.2148	0.1555	179	0.1480	0.2570	509	0.2148	0.1555
	04 Apr	509	0.1781	0.1578	175	0.1231	0.2527	509	0.1781	0.1578
	05 May	509	0.2187	0.2315	169	0.0854	0.1800	509	0.2187	0.2315
	06 Jun	509	-0.0251	0.0063	127	-0.0173	-0.0101	509	-0.0251	0.0063
	07 Jul	509	0.0096	0.0891	137	-0.2345	-0.1982	509	0.0096	0.0891
	08 Aug	509	0.0934	0.1624	118	-0.1358	-0.0990	509	0.0934	0.1624
	09 Sep	510	0.0433	0.0420	98	-0.0575	0.0727	510	0.0433	0.0420
	10 Oct	510	0.3157	0.3088	107	0.0420	0.1031	360	0.1609	0.2586
	11 Nov	510	0.1869	0.1204	4	-0.0826	0.4000	134	-0.0009	0.1652
	12 Dec	512	0.4209	0.3113	0	–	–	142	0.2020	0.2442
Altitude	<400 m	672	-0.0099	0.0280	18	-0.1147	-0.1084	451	-0.2279	-0.1031
	400–<800 m	2,461	0.0253	0.0679	222	-0.1328	-0.1002	1,721	-0.2877	-0.1542
	800–<1200 m	816	-0.0538	0.0545	109	-0.1565	-0.1091	620	-0.4003	-0.2356
	≥1200 m	2,165	-0.1242	-0.0076	873	-0.3248	-0.2453	1,894	-0.3803	-0.2882
Geography	North	1,261	-0.1543	-0.0968	144	-0.3181	-0.2773	902	-0.5031	-0.3555
	South	912	0.0467	0.1750	198	-0.3547	-0.2724	725	-0.2985	-0.1666
	East	1,877	0.0723	0.2192	508	-0.2830	-0.1625	1,455	-0.2697	-0.1056
	West	2,064	-0.0680	0.0309	372	-0.1886	-0.1001	1,604	-0.3479	-0.2470

Table 9: Regression results for the wind–solar relationship by geographical zone

Temporal scale	Model	Zone	Coefficient	Standard error	p-value
1. Hourly	Simple	North	7.3522	5.1234	1.5128e-01
		South	14.1971	6.1982	2.1991e-02
		West	6.3674	3.7379	8.8482e-02
		East	10.7395	3.9559	6.6315e-03
	Station FE	North	9.5663	6.1728	1.2120e-01
		South	18.2450	8.1525	2.5225e-02
		West	8.5833	5.1065	9.2789e-02
		East	12.8311	5.3767	1.7013e-02
	Time FE	North	0.0971	0.3355	7.7229e-01
		South	4.8450	0.9729	6.3661e-07
		West	1.4337	0.4152	5.5366e-04
		East	3.4087	0.6029	1.5727e-08
	Station + time FE	North	0.0671	0.4794	8.8876e-01
		South	5.6508	1.1528	9.4895e-07
		West	1.1053	0.5793	5.6380e-02
		East	2.1561	0.6173	4.7762e-04
2. Daily	Simple	North	-8.9242	1.2305	4.1100e-13
		South	1.9354	2.8243	4.9319e-01
		West	-4.1680	0.8998	3.6215e-06
		East	0.8026	1.4888	5.8983e-01
	Station FE	North	-13.6621	1.8426	1.2257e-13
		South	1.7499	4.6915	7.0915e-01
		West	-9.6085	1.6934	1.3965e-08
		East	-3.7039	2.4412	1.2921e-01
	Time FE	North	0.2595	0.4084	5.2514e-01
		South	4.7154	1.1160	2.3865e-05
		West	1.1119	0.5175	3.1650e-02
		East	4.0724	0.7777	1.6409e-07
	Station + time FE	North	0.1980	0.6658	7.6614e-01
		South	6.2364	1.4114	9.9406e-06
		West	0.1899	0.5582	7.3366e-01
		East	2.2578	0.7780	3.7102e-03
3. Monthly	Simple	North	-11.0312	3.3376	9.5478e-04
		South	2.8684	4.0465	4.7844e-01
		West	-3.2094	1.7084	6.0347e-02
		East	3.2070	2.2461	1.5339e-01
	Station FE	North	-38.3589	9.8386	9.7715e-05
		South	5.2669	19.6462	7.8864e-01
		West	-24.8229	7.3995	7.9949e-04
		East	-8.6941	9.2917	3.4947e-01
	Time FE	North	-0.4438	0.6862	5.1778e-01
		South	3.1088	1.4876	3.6679e-02
		West	1.5400	0.9187	9.3727e-02
		East	5.3984	1.0682	4.4601e-07
	Station + time FE	North	-2.0719	2.2962	3.6694e-01
		South	5.8456	2.3956	1.4711e-02
		West	-0.6744	1.1654	5.6282e-01
		East	2.5934	1.2090	3.1985e-02

B Regression results

B.1 Overall specifications

B.2 Geographic specifications

C Descriptive statistics

C.1 Overall statistics by sample and temporal resolution

Table 10: Mean hourly, daily and monthly wind speed and solar radiation by sample.

Sample	Scale	N	Wind (m/s)	Solar (W/m^2)
FULL	Hourly	4,453,593	2.700	155.196
FULL	Daily	185,736	2.701	155.145
FULL	Monthly	6,114	2.705	154.931
AND	Hourly	604,689	5.187	459.561
AND	Daily	32,108	4.785	217.047
AND	Monthly	1,222	4.385	200.056
OR	Hourly	2,331,485	4.008	286.727
OR	Daily	137,539	3.122	190.424
OR	Monthly	4,686	2.977	184.209

Notes: N counts paired wind–radiation observations. Daily and monthly means are calculated before applying the sample filters. The source export labels the unfiltered sample "Normal"; it is called FULL in the main text.

Source: Own calculations using MeteoSwiss data; `code_output/descriptive_statistics_overall.tex`.

C.2 Hourly statistics by geographical zone

C.3 Hourly statistics by altitude

Table 11: Hourly mean wind speed and solar radiation by geographical zone and sample.

Sample	**Zone**	**N**	**Wind (m/s)**	**Solar (W/m^2)**
FULL	North	919,029	2.348	147.379
FULL	South	665,049	2.531	162.432
FULL	East	1,364,806	2.885	155.382
FULL	West	1,504,709	2.822	156.604
AND	North	96,573	4.646	415.717
AND	South	83,577	5.139	499.243
AND	East	210,337	5.423	470.442
AND	West	214,202	5.219	453.159
OR	North	461,934	3.451	281.637
OR	South	335,988	3.752	311.103
OR	East	725,717	4.333	283.085
OR	West	807,846	4.141	282.771

Source: Own calculations using MeteoSwiss data;
`code_output/descriptive_statistics_by_geographic_zone.tex`.

Table 12: Hourly mean wind speed and solar radiation by altitude class and sample.

Sample	**Altitude**	**N**	**Wind (m/s)**	**Solar (W/m^2)**
Normal	< 400 m	490,610	1.788	152.241
Normal	400–< 800 m	1,794,627	2.188	150.139
Normal	800–$< 1,200$ m	595,071	2.432	149.225
Normal	$\geq 1,200$ m	1,573,285	3.670	164.146
AND	< 400 m	43,634	4.542	492.885
AND	400–< 800 m	205,930	4.831	461.720
AND	800–$< 1,200$ m	75,620	4.626	479.400
AND	$\geq 1,200$ m	279,505	5.703	447.399
OR	< 400 m	202,984	2.790	352.166
OR	400–< 800 m	825,872	3.307	313.478
OR	800–$< 1,200$ m	298,581	3.677	286.568
OR	$\geq 1,200$ m	1,004,048	4.929	251.541

Source: Own calculations using MeteoSwiss data;
`code_output/descriptive_statistics_by_altitude.tex`.

List of Figures

1 Geographical distribution of Swiss Weather Stations 3
2 Monthly profiles of mean daily wind speed and solar radiation, 2020–2023. 4
3 Hourly wind speed and solar radiation in the *FULL* sample ($N = 4{,}453{,}593$ station-hours), 2020–2023. Dashed lines indicate the thresholds of 3 m/s and 100 W/m^2. 7
4 Hourly wind speed and solar radiation in the *OR* sample ($N = 2{,}331{,}485$ station-hours), 2020–2023. Percentages are conditional on OR selection; the empty low-wind, low-radiation zone follows from the filter. 8
5 Monthly mean daily wind speed by elevation class and sample definition. . 10
6 Hourly wind–solar correlations by geographical zone, elevation class and sample definition. Cell colours and the first number show Spearman correlations; Pearson correlations are in parentheses. All panels use a common colour scale. Empty cells indicate unavailable correlations. 11

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